Total No. of Questions : 6]

SEAT No. :

P270

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B.E./INSEM/APR-597

B.E. (IT) (Semester - II)

414462 : DISTRIBUTED COMPUTING SYSTEM

(2015 Pattern)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.***
2) ***Figures to the right side indicate full marks.***
3) ***Assume suitable data if necessary.***

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Q1) a) Explain the concepts of middleware in distributed system. How it deals with Heterogeneity? **[5]**

b) What is a three - tiered client - server architecture? **[5]**

OR

Q2) a) Explain various design issues of distributed system. **[5]**

b) Explain what is meant by (distribution) transparency and give examples of different types of transparency. **[5]**

Q3) a) Define : **[4]**

i) Persistent Communication.

ii) Transient Communication.

iii) Asynchronous Communication.

iv) Synchronous Communication.

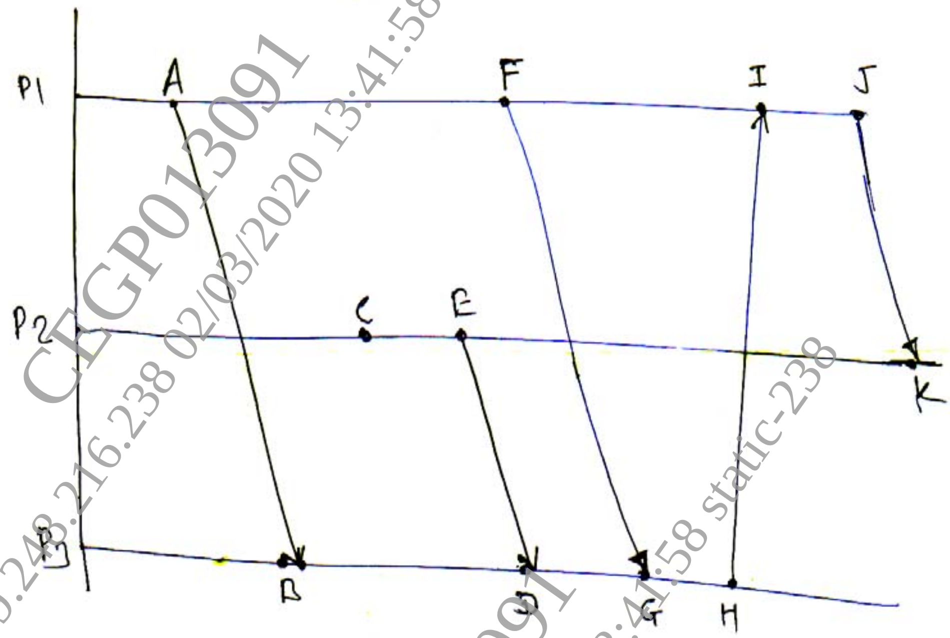
b) What are the drawback of Lamport's logical clock algorithms and how they are overcome in vector clock algorithm. **[6]**

OR

P.T.O.

Q4) a) State the differences between Ring and Bully Election algorithms. **[4]**

b) Solve following timing diagram using Vector - Time - Stamp Method. **[6]**



Q5) a) Show that Byzantine agreement cannot always be reached among four processors. Or two processors are fault. **[6]**

b) Distinguish between : **[4]**

i) Buffering and Caching.

ii) Caching and Replication.

OR

Q6) a) How failure is masked using redundancy? What is k-fault Tolerance system? **[6]**

b) Explain data-centric consistency models with its advantages and disadvantages. **[4]**

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